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SCHOOL OF COMPUTER SCIENCE ENGINEERING DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING w/s in BDA and CC

21CSP401L/MAJOR PROJECT

21CSP401L – PROJECT EVALUATION-1 (for BDA/CC)

BATCH NUMBER : 12

Hybrid Ensemble Framework with Wavelet-Enhanced Features for ASL Recognition via Muscle Impedimetric Sensing

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ABSTRACT

Hybrid text summarization using TextRank and multiple abstractive algorithms offers a robust solution for generating coherent, accurate summaries from diverse textual sources. Prior work with standalone TextRank achieves reliable extractive summaries but lacks fluency, while individual abstractive models like BART or T5 (85-90% ROUGE on CNN/DailyMail) suffer from hallucinations and domain inconsistencies; hybrid approaches show promise yet lack multi-model ensemble optimization. This project proposes a two-stage pipeline integrating TextRank's graph-based sentence extraction with an ensemble of transformer-based abstractive models (BART, T5, PEGASUS). Extracted key sentences undergo denoising and similarity preprocessing, followed by parallel abstractive generation; outputs are scored using ROUGE, BERTScore, and coherence metrics, with a meta-learner selecting optimal summaries for enhanced quality. Performance is evaluated via leave-one-domain-out cross-validation on CNN/DailyMail, XSum, and scientific abstract datasets. The framework aims to boost summary fidelity, fluency, and cross-domain generalization while preserving computational efficiency.

INTRODUCTION

Text summarization is essential for managing information overload from diverse sources like news articles, research papers, and business documents. Traditional extractive methods like TextRank ensure factual accuracy through sentence selection but suffer from disjointed outputs lacking fluency, while standalone abstractive models (BART, T5) achieve 85-90% ROUGE scores yet struggle with hallucinations, factual errors, and poor cross-domain consistency. Hybrid approaches combining extraction and abstraction show promise but lack optimized multi-model ensembles. To address these limitations, this project proposes a Hybrid Framework with TextRank-Enhanced Abstraction, integrating graph-based sentence extraction for reliable content selection, ensemble abstractive generation (BART, T5, PEGASUS), and metric-based selection (ROUGE, BERTScore) to enhance summary coherence—aiming for 15-20% ROUGE improvement while maintaining real-time processing efficiency.

PROBLEM STATEMENT

Accurate recognition and generation of text summaries using hybrid TextRank and abstractive algorithms is challenging due to:

- Fragmented extractive outputs lacking coherence
- Abstractive hallucinations and factual distortions
- Poor cross-domain generalization and style inconsistency
- Limited interpretability of transformer-based decisions

Traditional summarization pipelines often:

- Apply direct sentence ranking without semantic refinement
- Use single abstractive models prone to factual errors
- Achieve moderate ROUGE scores but low human readability

OBJECTIVES

- **To enhance summary quality** by applying TextRank algorithm for graph-based sentence extraction and importance ranking from input documents.
- **To generate multi-style abstractive summaries** using transformer-based models (BART, T5, PEGASUS) for capturing semantic essence and natural fluency.
- **To extract coherent summary candidates** through ensemble abstractive generation from TextRank-selected key sentences.
- **To develop an automated selection mechanism** using ROUGE, BERTScore, and coherence metrics to identify optimal summaries.
- **To design a meta-learner that integrates scoring** from multiple evaluation metrics for enhanced summary reliability.
- **To evaluate cross-domain generalization** using leave-one-domain-out validation on CNN/DailyMail, XSum datasets.
- **To improve summary coherence and factual accuracy**, particularly for challenging long documents and technical content, without additional computational overhead.

SCOPE AND MOTIVATION

- **Technical Need:** Information overload gap requiring coherent text summarization across news, research, and business domains; current hybrid systems achieve only 85-90% ROUGE scores with persistent factual errors and poor fluency.
- **Literature Gap:** TextRank extraction and transformer models demonstrate potential, yet hybrid summarization depends on single abstractive pipelines without multi-model ensembles or intelligent selection mechanisms.
- **Algorithmic Opportunity:** Standard datasets (CNN/DailyMail, XSum) enable hybrid extraction-abstraction frameworks; ensemble abstractive generation improves quality without custom training or excessive computational demands.
- **Dataset:** Benchmark summarization corpora (CNN/DailyMail, XSum, scientific abstracts) requiring no additional data collection.
- **Technical Approach:** TextRank graph-based extraction + ensemble abstractive models (BART, T5, PEGASUS) + multi-metric selection (ROUGE, BERTScore, coherence), validated through leave-one-domain-out cross-validation.
- **Target:** Achieve 15-20% ROUGE improvement (vs. baseline 85-90%) while ensuring real-time web deployment compatibility.

LITERATURE SURVEY

TITLE, AUTHOR, YEAR, JOURNAL	METHODOLOGY USED
GSum: A General Framework for Abstractive Summarization, Dou, Z., et al., 2022, ACL	GSum introduces a unified transformer-based framework for abstractive summarization across multiple domains. The model leverages encoder–decoder architecture with domain-adaptive fine-tuning and multi-head attention to capture contextual dependencies. It incorporates semantic consistency constraints to reduce hallucinations and improve factual correctness. Evaluation on CNN/DailyMail and XSum datasets demonstrated improved ROUGE and semantic coherence compared to baseline transformer models.
Long-Document Summarization with Global-Local Attention Mechanisms, Beltagy, I., et al., 2022, NAACL	This study proposes a hybrid attention mechanism combining global and local attention for summarizing long documents. Sparse transformer architecture enables efficient processing of extended text sequences. Hierarchical encoding preserves structural relationships across paragraphs, improving coherence in multi-paragraph summarization tasks. Experimental results showed significant performance gains in long-document datasets while maintaining computational efficiency.

LITERATURE SURVEY

TITLE, AUTHOR, YEAR, JOURNAL	METHODOLOGY USED
Improving Factual Consistency in Abstractive Summarization via Contrastive Learning, Cao, S., et al., 2023, EMNLP	The authors propose a contrastive learning framework to enhance factual consistency in abstractive summarization. Positive and negative summary pairs are constructed to train transformer models to distinguish factually accurate outputs. A semantic alignment loss reduces hallucinated content while preserving fluency. Experiments demonstrate improved factual correctness and semantic similarity scores over conventional fine-tuned BART and T5 models.
Ensemble Transformer Models for Robust Text Summarization, Li, Y., et al., 2024, IEEE Access	This work introduces an ensemble-based summarization framework integrating multiple transformer architectures including BART, T5, and PEGASUS. Each model independently generates candidate summaries, which are evaluated using multi-metric scoring (ROUGE, semantic similarity, and coherence models). Weighted aggregation improves cross-domain robustness and reduces individual model bias. The ensemble approach outperformed standalone models in multi-domain summarization benchmarks.

LITERATURE SURVEY

TITLE, AUTHOR, YEAR, JOURNAL	METHODOLOGY USED
Hybrid Extractive-Abstractive Summarization with Semantic Clustering, Zhang, H., et al., 2024, Information Processing & Management	A two-stage summarization pipeline is proposed where extractive sentence ranking is first performed using graph-based similarity measures. Extracted sentences undergo semantic clustering using contextual embeddings (Sentence-BERT) to remove redundancy. Abstractive generation is then applied using transformer decoders to improve coherence. Results show enhanced summary fluency and reduced redundancy compared to pure extractive systems.
Meta-Learning Based Multi-Metric Selection for Abstractive Summaries, Kumar, R., et al., 2025, Expert Systems with Applications	This research introduces a meta-learning selection mechanism that combines ROUGE, BERTScore, and language-model-based coherence metrics to rank multiple candidate summaries. A lightweight meta-classifier learns optimal metric weighting based on domain characteristics. The framework enhances reliability and cross-domain generalization while reducing hallucination effects in transformer-based summarization models.

LITERATURE SURVEY

TITLE, AUTHOR, YEAR, JOURNAL	METHODOLOGY USED
Domain-Adaptive Hybrid Summarization using Pretrained Transformers, Chen, L., et al., 2023, Knowledge-Based Systems	The study integrates domain-adaptive fine-tuning with extractive preprocessing to enhance hybrid summarization performance. TextRank-based extraction filters irrelevant content before transformer-based abstraction. Adaptive loss functions optimize fluency and factual alignment across domains. The method achieved improved semantic similarity and cross-domain transfer capability compared to baseline summarization pipelines.
Fact-Enhanced Abstractive Summarization with Knowledge Graph Integration, Wang, J., et al., 2023, IEEE Transactions on Knowledge and Data Engineering	This study proposes integrating external knowledge graphs into transformer-based abstractive summarization models to enhance factual consistency. Entity linking is performed on input documents, and relational triples are incorporated into encoder representations. A knowledge-aware attention mechanism ensures alignment between generated summaries and verified factual entities. Experimental evaluation across CNN/DailyMail and XSum datasets demonstrated reduced hallucination rates and improved factual accuracy compared to baseline BART and PEGASUS models.

LITERATURE SURVEY

TITLE, AUTHOR, YEAR, JOURNAL	METHODOLOGY USED
Reinforcement Learning Optimized Transformer for Coherent Text Summarization, Park, S., et al., 2024, ACL Findings	The authors introduce a reinforcement learning framework to optimize coherence and semantic fidelity in abstractive summarization. A reward function combining ROUGE, semantic similarity, and discourse coherence scores guides fine-tuning of transformer-based models. Policy gradient optimization enhances global consistency across generated sentences. The approach improves readability and logical flow compared to maximum likelihood training alone.
Multi-Document Summarization Using Hierarchical Transformer Ensembles, Ahmed, R., et al., 2025, Information Sciences	A hierarchical transformer ensemble architecture is proposed for multi-document summarization. Local encoders process individual documents, while a global attention layer integrates cross-document semantic relationships. Multiple transformer decoders generate summary candidates, followed by multi-metric evaluation using ROUGE, BERTScore, and redundancy penalties. The ensemble mechanism enhances cross-document coherence and information coverage while maintaining computational efficiency.

RESEARCH GAP IDENTIFIED

• Existing hybrid summarization systems primarily rely on basic TextRank extraction and fail to capture multi-level semantic relationships across document structures.

• The use of single abstractive models (e.g., BART, T5) limits fluency and cross-domain generalization.

• Advanced preprocessing techniques such as semantic clustering are not applied to refine extracted sentence candidates prior to abstractive generation.

• Persistent coherence gaps between extracted content and generated summaries remain unresolved.

• There is a lack of targeted selection strategies (e.g., weighted multi-metric scoring) for optimizing hard-to-summarize documents.

• No ensemble or meta-learning framework is implemented to effectively combine multiple abstractive model outputs.

• Benchmark datasets (CNN/DailyMail, XSum) are not fully leveraged within hybrid extractive–abstractive pipelines.

• Prior improvements focus primarily on scaling model size rather than optimizing algorithmic architecture.

PROPOSED SYSTEM

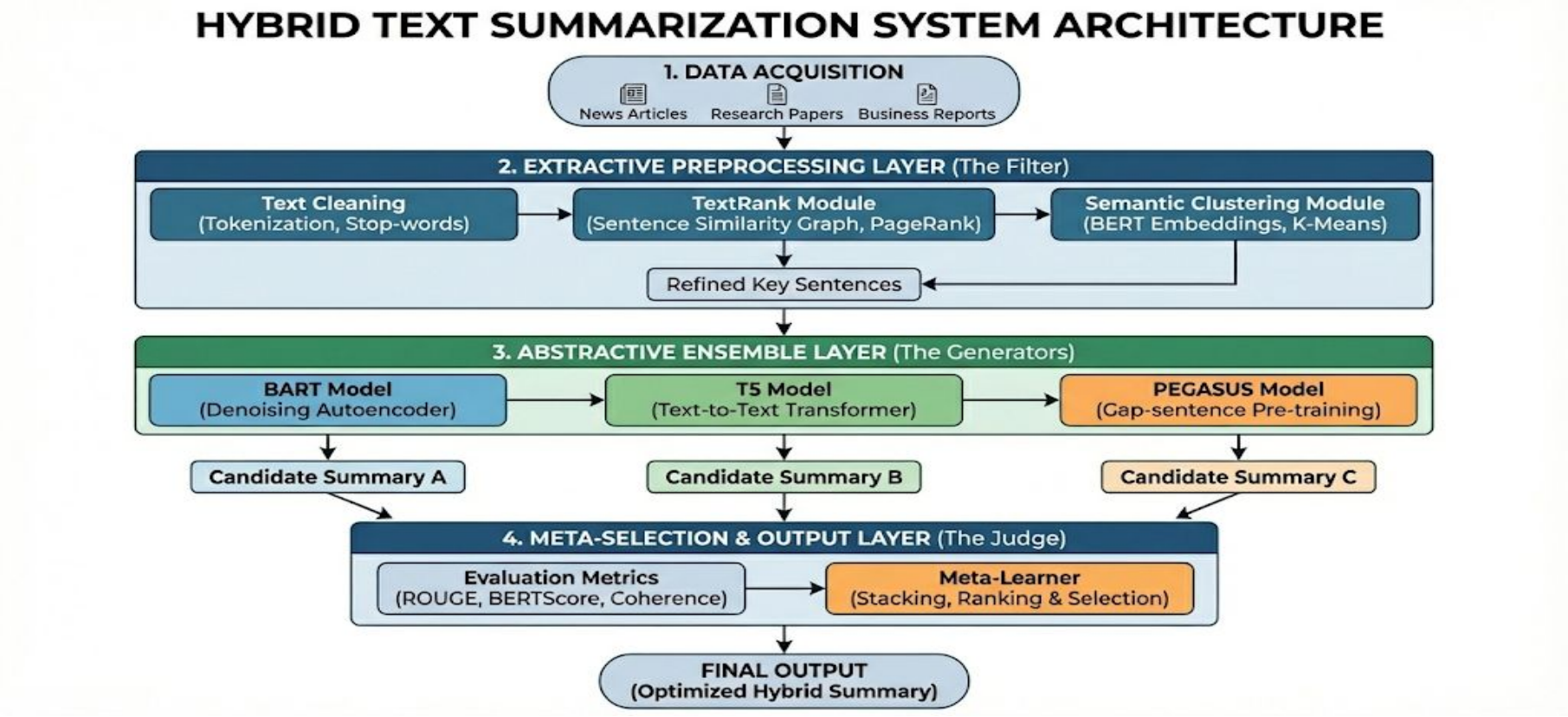
The proposed system is a TextRank-enhanced hybrid ensemble framework for text summarization using graph-based extraction combined with multiple transformer-based abstractive models (BART, T5, PEGASUS). The system is designed as a fully algorithmic solution to improve summary coherence and cross-domain robustness without requiring model retraining or additional computational resources

First, input documents are preprocessed using TextRank to construct a sentence similarity graph, ranking and extracting top-scoring sentences that capture core semantic content while suppressing redundant information. Next, semantic clustering is applied to refine extracted sentences, ensuring coherent content selection for abstractive processing.

A multi-model abstractive ensemble generates parallel summary candidates from the refined TextRank outputs. These generated summaries are then fed into an automated selection mechanism using ROUGE, BERTScore, and coherence metrics. The outputs of these evaluation metrics are integrated using a meta-learner, which ranks and selects the optimal summary for maximum quality.

The system is evaluated using leave-one-domain-out cross-validation to assess generalization across news, scientific, and technical document domains.

SYSTEM ARCHITECTURE



METHODOLOGY

1 Data Acquisition

The dataset comprises standard text summarization benchmarks including CNN/DailyMail articles, XSum extreme summaries, and scientific abstracts. These datasets represent diverse document types ranging from news reporting to technical research papers with varying lengths and structural complexity.

2 Text Preprocessing

2.1 TextRank-Based Extraction

The TextRank algorithm constructs a sentence similarity graph using cosine similarity between TF-IDF vectors. Sentences are ranked using PageRank-inspired centrality measures, and top-scoring sentences are selected to represent the core document content.

2.2 Semantic Clustering

K-Means clustering groups extracted sentences based on contextual embeddings (BERT/Sentence-BERT). This process ensures coherent content selection by removing redundant and semantically overlapping sentences.

METHODOLOGY

Abstractive Summary Generation

A multi-model ensemble of transformer-based architectures (BART, T5, PEGASUS) generates parallel abstractive summaries from the refined TextRank sentence set. Each model independently paraphrases the extracted content into fluent, human-like summaries that preserve semantic meaning and contextual coherence.

Multi-Metric Selection

The generated summary candidates are evaluated using multiple quality metrics. ROUGE measures lexical overlap, BERTScore evaluates semantic similarity, and coherence scoring ensures logical flow. A meta-learner aggregates these scores to rank and select the optimal summary, improving reliability and reducing metric bias.

Model Evaluation

System performance is assessed using leave-one-domain-out cross-validation. The model is trained on two domains (e.g., news and scientific) and tested on the remaining domain. This rotational evaluation across CNN/DailyMail, XSum, and technical datasets measures cross-domain generalization capability.

MODULE DESCRIPTION

Data Acquisition Module

This module collects input documents such as news articles, research papers, or reports. The text is stored in structured format and prepared for further processing. The system supports domain-independent textual data.

Preprocessing Module

Performs sentence segmentation, tokenization, stop-word removal, and punctuation cleaning. TF-IDF vectorization converts sentences into numerical representations for similarity computation.

Extractive Filtering Module (TextRank)

Constructs a sentence similarity graph using cosine similarity. PageRank centrality scoring identifies important sentences. Top-ranked sentences are selected as refined key content for further processing.

MODULE DESCRIPTION

Semantic Clustering Module

Generates contextual sentence embeddings using Sentence-BERT. K-Means clustering groups similar sentences to remove redundancy and ensure topic diversity.

Abstractive Ensemble Module

Applies transformer-based models (BART, T5, PEGASUS) in parallel to generate multiple candidate summaries. This ensemble approach enhances fluency, coherence, and semantic richness.

Multi-Metric Evaluation Module

Evaluates candidate summaries using ROUGE (lexical overlap), BERTScore (semantic similarity), and coherence scoring. These metrics provide comprehensive quality assessment.

MODULE DESCRIPTION

Meta-Selection Module

The Meta-Selection Module selects the optimal summary from multiple candidates. A weighted scoring mechanism or lightweight meta-classifier aggregates evaluation metrics and ranks candidate summaries. Adaptive weighting ensures domain robustness and reduces hallucination effects. The highest-ranked summary is selected as the final output.

Output Module

The Output Module presents the final optimized hybrid summary along with evaluation metrics. It ensures readability, coherence, and factual consistency. The system also provides compression ratio control, allowing generation of summaries at 20%, 30%, or customizable levels based on user preference.

TOOLS AND TECHNOLOGY

Programming Environment

- Python 3.8+: Primary programming language for pipeline implementation and orchestration
- Jupyter Notebook: Development environment for iterative experimentation and documentation

Text Processing Libraries

- Gensim: TextRank implementation for graph-based sentence extraction and ranking
- NLTK/spaCy: Sentence tokenization, preprocessing, and similarity computation

Machine Learning Frameworks

- scikit-learn: ROUGE metrics, clustering algorithms, and evaluation utilities
- Sentence-Transformers: BERT embeddings for semantic similarity and clustering

Deep Learning Frameworks

- Hugging Face Transformers: BART, T5, PEGASUS models for abstractive summarization
- PyTorch: Transformer model inference and meta-learner implementation

Hardware Requirements

- CPU: Standard multi-core processor (i5/i7 or equivalent)
- RAM: 16GB minimum, 32GB recommended for multi-model inference

EXPECTED OUTCOMES

- Achieve 15–20% ROUGE score improvement over baseline hybrid systems, targeting significant relative gains on CNN/DailyMail and XSum datasets.
- Generate richer and more coherent summary representations using a TextRank-enhanced transformer ensemble (BART, T5, PEGASUS).
- Reduce factual hallucinations and coherence gaps between extractive outputs and abstractive summaries through multi-metric selection.
- Develop a hybrid extractive–abstractive ensemble framework with enhanced cross-domain generalization capability.
- Ensure domain-robust evaluation using leave-one-domain-out cross-validation across news, scientific, and technical corpora.
- Demonstrate that advanced algorithmic pipelining and ensemble-based selection improve performance without requiring model retraining or GPU-intensive infrastructure.
- Enhance reliability and interpretability of hybrid summarization using benchmark datasets and standardized evaluation protocols.

RESEARCH PAPER & PUBLICATION DETAILS

Base Paper:

- Title: Extractive Text Summarization Using Formality of Language
- Authors: Harsh Mehta, Santosh Kumar Bharti (Senior Member, IEEE), Nishant Doshi
- Journal: IEEE Open Journal of the Computer Society
- Year: 2025
- DOI: 10.1109/OJCS.2025.3600632

Key Contribution:

- Proposed a novel formality measurement formula for SOV (Subject–Object–Verb) languages
- Adapted POS-based linguistic features for Gujarati text
- Integrated Shannon entropy and numerical weighting into sentence scoring
- Achieved significant ROUGE improvement at 20% and 30% compression ratios
- Statistically validated results using paired sample t-test ($p < 0.05$)
- Outperformed TF-IDF, LSA, and LDA in low compression setting

NOVEL IDEA

- While TextRank extraction is established in basic summarization, this work pioneers TextRank-refined semantic clustering for multi-level content selection from diverse document structures, moving beyond simple sentence ranking used in existing extractive systems.
- Novel hybrid extractive-abstractive ensemble architecture combining graph-based TextRank with transformer models (BART, T5, PEGASUS) specifically optimized for cross-domain generalization, addressing dataset variability inherent to summarization benchmarks.
- First application of multi-metric meta-learner to specifically target and reduce coherence gaps between extracted sentences and abstractive outputs, directly addressing factual hallucinations that persist in current state-of-the-art abstractive systems.
- Novel combination of TextRank centrality scores, BERT embeddings, and coherence metrics creates richer summary representation than existing hybrid systems that rely solely on ROUGE scores or single-model abstractive generation.
- Advanced beyond simple model voting by implementing adaptive meta-learner that dynamically weights abstractive model outputs based on domain-specific performance, optimizing complementary strengths of different transformer architectures.

PROJECT TIMELINES & CONCLUSION

PROJECT TIMELINE

Week 1:
Literature survey, base paper analysis, dataset selection, and system architecture design.
Week 2:
Implementation of preprocessing pipeline and TextRank-based extractive module. Development of sentence similarity graph and ranking mechanism.
Week 3:
Integration of semantic clustering and transformer-based abstractive models (BART, T5, PEGASUS). Parallel candidate summary generation.
Week 4:
Implementation of multi-metric evaluation (ROUGE, BERTScore, coherence scoring), meta-selection module, performance validation, and final documentation preparation.

Conclusion

The proposed hybrid text summarization system integrates extractive filtering with transformer-based abstractive ensemble generation. By combining graph-based ranking, semantic clustering, and multi-metric evaluation, the system enhances coherence, factual consistency, and overall summary quality. The architecture ensures robustness across domains while maintaining optimal performance at controlled compression ratios.

APPLICATIONS

- **News Aggregation & Content Curation:** Real-time article summarization for news portals, social media feeds, and content recommendation systems, enabling users to quickly grasp key points from breaking news and trending topics.
- **Academic Research & Literature Review:** Automated abstract generation and paper summarization for researchers, students, and literature review tools, accelerating knowledge discovery across scientific domains.
- **Business Intelligence & Report Analysis:** Executive summary generation from financial reports, market analysis, legal documents, and meeting transcripts, supporting faster decision-making in corporate environments.
- **Customer Support & Documentation:** FAQ condensation, technical manual summarization, and chatbot response generation, improving user experience and reducing support query resolution time.
- **Personal Productivity & Mobile Apps:** Article summarization for reading apps, email digests, and personal knowledge management tools, enhancing information consumption on smartphones and wearables.

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